

DBSE & DBD WEEK 2 PRACTICAL

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ROLL NO:2520030557

SECTION:8

```
Microsoft Windows [Version 10.0.26200.8875]
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C:\Users\ggott>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>MySQL -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE Books (
  ->     book_id INT PRIMARY KEY,
  ->     title VARCHAR(100) NOT NULL,
  ->     isbn VARCHAR(20) UNIQUE,
  ->     published_year INT CHECK (published_year < 2027)
  -> );
ERROR 1050 (42S01): Table 'books' already exists
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
  -> (1, 'The Great Gatsby', '9780743273565', 1925),
  -> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
  -> (3, '1984', '9780451524935', 1949);
ERROR 1062 (23000): Duplicate entry '1' for key 'books.PRIMARY'
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title      | isbn      | published_year |
+-----+-----+-----+-----+
| 1       | JAVA      | 9780061122415 | 1988          |
```

book_id	title	isbn	published_year
1	JAVA	9780061122415	1988
2	DBMS	9780132350884	2008
3	ML	9780735211292	2018
4	Animal Farm	9780451526342	1945

4 rows in set (0.00 sec)

```
mysql> CREATE TABLE Members (
->     member_id INT PRIMARY KEY,
->     full_name VARCHAR(100),
->     email VARCHAR(100) UNIQUE
-> );
ERROR 1050 (42S01): Table 'members' already exists
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
-> (101, 'John Smith', 'john.smith@email.com'),
-> (102, 'Emma Wilson', 'emma.wilson@email.com'),
-> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Members;
```

member_id	full_name	email
1	SUBBU ROY	subbusir@example.com
2	SATEESH	sateesh@example.com
3	KRISHNA	krishna@example.com
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

6 rows in set (0.00 sec)

6 rows in set (0.00 sec)

```
mysql> CREATE TABLE Loans (  
->     loan_id INT PRIMARY KEY,  
->     member_id INT,  
->     book_id INT,  
->     loan_date DATE,  
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),  
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)  
-> );
```

ERROR 1050 (42S01): Table 'loans' already exists

```
mysql> CREATE TABLE Loans (  
->     loan_id INT PRIMARY KEY,  
->     member_id INT,  
->     book_id INT,  
->     loan_date DATE,  
->     FOREIGN KEY (member_id) REFERENCES Members(member_id),  
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)  
-> );
```

ERROR 1050 (42S01): Table 'loans' already exists

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES  
-> (1, 101, 1, '2025-01-05'),  
-> (2, 102, 2, '2025-01-08'),  
-> (3, 103, 3, '2025-01-10'),  
-> (4, 101, 2, '2025-02-01'),  
-> (5, 102, 1, '2025-02-05'),  
-> (6, 103, 2, '2025-02-12'),  
-> (7, 101, 3, '2025-03-01'),  
-> (8, 102, 3, '2025-03-07'),  
-> (9, 103, 1, '2025-03-15'),  
-> (10, 101, 1, '2025-04-01');
```

Query OK, 10 rows affected (0.03 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Loans;
```

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT m.full_name AS Member_Name,  
->      b.title AS Book_Title  
-> FROM Loans l  
-> INNER JOIN Members m  
-> ON l.member_id = m.member_id  
-> INNER JOIN Books b  
-> ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	JAVA
Emma Wilson	JAVA
Michael Brown	JAVA
John Smith	JAVA
Emma Wilson	DBMS
John Smith	DBMS
Michael Brown	DBMS
Michael Brown	ML
John Smith	ML
Emma Wilson	ML

```

10 rows in set (0.01 sec)

mysql> SELECT published_year,
->         COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
+-----+-----+
| published_year | Total_Books |
+-----+-----+
|          1945 |          1 |
|          1988 |          1 |
|          2008 |          1 |
|          2018 |          1 |
+-----+-----+
4 rows in set (0.00 sec)

mysql> CREATE TABLE Donation_History (
->     donation_id INT PRIMARY KEY,
->     book_id INT,
->     donor_name VARCHAR(100),
->     donation_date DATE,
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
ERROR 1050 (42S01): Table 'donation_history' already exists
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
ERROR 1062 (23000): Duplicate entry '4' for key 'books.PRIMARY'
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
-> VALUES (1, 4, 'Raj Kumar', CURDATE());
ERROR 1062 (23000): Duplicate entry '1' for key 'donation_history.PRIMARY'
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM Donation_History;

```

```
mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
|          1 |        4 | Raj Kumar  | 2026-07-29    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> CREATE INDEX idx_books_isbn
      -> ON Books(isbn);
ERROR 1061 (42000): Duplicate key name 'idx_books_isbn'
mysql> SELECT * FROM Books
      -> WHERE isbn = '9780451524935';
Empty set (0.00 sec)
```

```
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title      | isbn          | published_year |
+-----+-----+-----+-----+
|        1 | JAVA      | 9780061122415 | 1988           |
|        2 | DBMS      | 9780132350884 | 2008           |
|        3 | ML        | 9780735211292 | 2018           |
|        4 | Animal Farm | 9780451526342 | 1945           |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name    | email                      |
+-----+-----+-----+
|          1 | SUBBU ROY    | subbusir@example.com      |
|          2 | SATEESH      | sateesh@example.com       |
|          3 | KRISHNA      | krishna@example.com       |
|        101 | John Smith   | john.smith@email.com      |
|        102 | Emma Wilson  | emma.wilson@email.com     |
|        103 | Michael Brown | michael.brown@email.com   |
+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Donation_History;
```

donation_id	book_id	donor_name	donation_date
1	4	Raj Kumar	2026-07-29

```
1 row in set (0.00 sec)
```

```
mysql> SHOW TABLES;
```

Tables_in_bookflow_db
books
donation_history
loans
members

```
4 rows in set (0.01 sec)
```

```
mysql> |
```